

Directed Numbers (Positive & Negative)

WorksheetHub AI
Mathematics · Grade 7

Unit 1 · Number

Easy

Directed numbers

Name: _____

Date: _____

Score: / 23



Remember

Positive numbers (+) are above zero. Negative numbers (−) are below zero.

Adding a negative is the same as subtracting: $5 + (-2) = 5 - 2 = 3$.

Subtracting a negative is the same as adding: $5 - (-2) = 5 + 2 = 7$.

start at −3, move +5, land on 2



Adding moves RIGHT →, subtracting moves LEFT ←

Multiply / divide

same signs →
positive

$$(-) \times (-) = +, (+) \times (+) = +$$

Multiply / divide

different signs →
negative

$$(-) \times (+) = -, (+) \times (-) = -$$

Tip

Use the number line or think of money: owing £5 is −5.

A. Add and subtract

1 $-5 + 8 = ?$

Answer:

2 $4 - 9 = ?$

Answer:

3 $-6 - (-2) = ?$

Answer:

4 $-3 + (-7) = ?$

Answer:

5 $-10 + 10 = ?$

Answer:

6 $2 - (-6) = ?$

Answer:

7 $-9 + 4 = ?$

Answer:

8 $-1 - 5 = ?$

Answer:

B. Multiply and divide

9 $-4 \times 6 = ?$

Answer:

10 $-8 \times -3 = ?$

Answer:

11 $7 \times -5 = ?$

Answer:

12 $24 \div -6 = ?$

Answer:

13 $-45 \div -9 = ?$

Answer:

14 $-36 \div 4 = ?$

Answer:

C. Order of operations

15 $3 + (-2) \times 4 = ?$

Answer:

16 $(-5 + 2) \times (-3) = ?$

Answer:

17 $10 - 3 \times (-2) = ?$

Answer:

D. Real-life problems

- 18 The temperature at night is -4°C . By midday it rises by 11°C . What is the temperature at midday?

Answer:

- 19 A lift starts in the basement at level -3 and goes up 8 floors. Which level does it reach?

Answer:

- 20 Sam has \$20 and owes his friend \$35. What is his total money? Write it as a directed number.

Answer:

- 21 A submarine at -120 m dives a further 45 m . How deep is it now?

Answer:

★ Challenge

- 22 Find two numbers that multiply to give -12 and add to give 1 .

Answer:

- 23 The temperature on Monday was -3°C . On Tuesday it was 5 degrees colder, on Wednesday 9 degrees warmer. What was Wednesday's temperature?

Answer: